

SUMMARY

Senior data scientist building production ML, multi-agent AI, and decision systems from problem framing through deployment and monitoring. Ph.D. researcher and open-source ML author whose software is used by 100+ research labs.

EXPERIENCE

Huntington National Bank · Senior Data Scientist

2023 – Present

- Agentic AI.** Lead a four-layer multi-agent architecture spanning specialist agents, agent-human discussion, orchestration and reconciliation, and verification against source data; powers customer synthesis, grounded content generation, and self-service prospecting without SQL.
- Intelligent applications.** Built decision applications using entity resolution, fuzzy matching, dynamic calculations, forward-looking segmentation, and executive outreach planning; took products from prototype through scheduled production delivery.
- Production ML & reliability.** Built and productionized recommendation models across commercial-banking product areas and led the ranking layer that converts model outputs into prioritized recommendations. Designed monitoring across three failure points: data quality, model quality via AUC, and decision usefulness via downstream lift.
- Technical leadership.** Lead team priorities with product and sales stakeholders, extend ML frameworks into new product areas, and present technical roadmaps to regional leadership.

Carnegie Mellon University · Ph.D. Researcher, Neuroscience

2018 – 2023

- B-SOiD.** Created an open-source unsupervised behavior-discovery pipeline and no-code application; published in *Nature Communications* and adopted by 100+ labs / 250+ users.
- A-SOiD.** Co-first-authored an active-learning platform that surpassed all 49 Alcrowd challenge entries using 12% of the training data; published in *Nature Methods*.
- Neural decoding.** Built a 24/7 recording system and cloud pipeline processing 160 TB of synchronized video and neural data to decode behavior directly from neural activity.

PERSONAL PROJECT

Vowel for Weddings · Full-Stack iOS App

2026

Designed, built, and shipped a native SwiftUI app with a Supabase backend, grounded LLM assistant, face matching for photo discovery, and push notifications.

SKILLS

Machine learning	Gradient boosting (LightGBM, XGBoost, scikit-learn) · deep learning (PyTorch, TensorFlow / Keras) · learning-to-rank & recommender systems · propensity modeling · active learning · clustering (UMAP, HDBSCAN) · evaluation (AUC, lift, calibration)
Applied AI	Amazon Bedrock · multi-agent orchestration · RAG / grounding · human-in-the-loop systems · verification & structured evaluation
Production ML	AWS SageMaker · Snowflake · scheduled AWS pipelines · Shakudo · Evidently AI · Anomalo · model monitoring · model governance & validation (MRM)
Applications & data	Python · SQL · Polars / Pandas · FastAPI · React · Streamlit · Power BI · Tableau · Swift · Supabase · Fly.io · Git · Linux

EDUCATION

- Ph.D., Neuroscience

Carnegie Mellon University · 2023

Presidential Fellowship · \$50K Oracle for Research grant
- M.S.

Boston University School of Medicine · 2016
- B.S.

University of Illinois at Urbana-Champaign · 2013

SELECTED PUBLICATIONS

- Schweihoff J.F., **Hsu A.I.**, et al. "A-SOiD, an active-learning platform for expert-guided, data-efficient discovery of behavior." *Nature Methods*, 2024. Co-first author.
- Hsu A.I.**, Yttri E.A. "B-SOiD, an open-source unsupervised algorithm for identification and fast prediction of behaviors." *Nature Communications*, 2021.